

EEGLAB For ELECTROPHYSIOLOGICAL SIGNAL PROCESSING

JAI SACHITH PAUL

EEGLAB is a GNU general-public-licensed MATLAB toolbox for processing electrophysiological data from electroencephalography (EEG), magnetoencephalography (MEG) and others. Users can perform tasks like independent component analysis (ICA), time/frequency analysis (TFA), artefact rejection and several modes of data visualisation.

The toolbox, called ICA/EEG toolbox in 1997, has travelled a long way from its initial release on the Internet as a set of data-processing functions. This was done by Scott Makeig in Computational Neurobiology Laboratory under the direction of Terry Sejnowski at Salk Institute, USA. Graphical interface along with artefact-removal functions were added by Arnaud Delorme, and the first stable and fully-documented version of EEGLAB was made in 2003. A number of biomedical engineers and method developers worldwide have made effective use of the tool since then.

What the software is capable of

EEGLAB toolbox contains more than 380 stand-alone MATLAB functions, over 50,000 lines of code and about 20 user-contributed plugins.

Data analysis and processing.

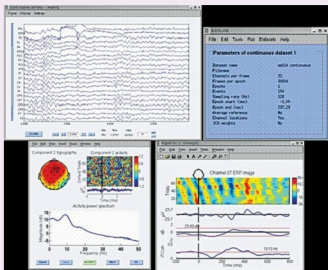
Users can input data in about 20 binary file formats and involve some amount of data pre-processing. There is an interactive graphical interface that allows processing of

System requirements

Operating system: Linux/Unix, Windows, Mac OSX or others capable of running MATLAB

Software: MATLAB 7 or later. Some latest toolboxes require object-oriented programming capabilities of MATLAB 7.6 or later

Recommended MATLAB toolboxes: Signal-processing toolbox, statistics toolbox (depending upon applications). Normally, just basic MATLAB licence suffices



EEGLAB window and its plots

high-density biomedical data using techniques composed of ICA, TFA and other standard averaging methods. Artefactual ICA components can be eliminated from the data.

Detailed analysis and processing can be carried out on ICA components representing activities of the brain. Tutorial and Help windows along with command history function allow users to run batch or custom data-analysis scripts.

Data visualisation. If users

try to model event-driven dynamics of the brain or visualise processed data for better analysis, the toolkit provides the necessary offerings. Analysis of data as an individual dataset or as a study-set containing multiple datasets could be done with the help of EEGLAB.

Who can make use of the toolkit

Hundreds of programming experts and researchers make use of EEGLAB for biomedical applications.

An expert MATLAB user.

EEGLAB offers a well-structured programming environment, which enables an experienced MATLAB user to play around with EEG signals. The user can store, access, measure, manipulate and visualise event-related EEG data. He or she